CRISP: Conformal Resilient Intrusion Detection with Selective Prediction

A. Moccardi, A. Amato, C. Ghosh, M. Cirillo

***Abstract*—Current deep learning classifiers for IoT network intrusion detection achieve high accuracy but provide no formal guarantee that any single prediction is correct. They cannot defer ambiguous inputs to a human analyst, and they remain vulnerable to adversarial perturbations that cause silent misclassification. We propose CRISP, a post-hoc framework that wraps any pre-trained classifier with Mondrian conformal prediction—a statistical method that replaces single-label outputs with prediction *sets* whose coverage probability is mathematically guaranteed for each traffic class independently. The size of each set drives an operational decision: a single label means the system is confident (predict), multiple labels signal ambiguity (abstain and defer), and an empty set flags a potential attack or unknown input (alarm). Within the same framework, a class-conditional score detects adversarial inputs without requiring a separate detection module. We evaluate CRISP on two large-scale IoT benchmarks—ACI-IoT-2023 (1.2M flows, 11 classes) and CICIDS2017 (2.1M flows, 8 classes)—using a 1D-CNN and an Attention-DNN as base classifiers. Across twenty experimental configurations, per-class coverage deviates from the theoretical target by at most $\pm$0.25%. At significance level α=0.01, the system classifies 93% of traffic automatically with 99.9% accuracy, deferring the rest, at under 4 μs per-sample overhead. The adversarial detector reaches ROC-AUC 0.87 under projected gradient descent (PGD) attacks—the only tested method that provides detection within a coverage-guaranteed pipeline. A reversal in detection effectiveness between the two architectures emerges as attack strength increases: the CNN detects weak perturbations more effectively (AUC 0.80 vs. 0.51), while the Attention-DNN dominates under strong iterative attacks (AUC 0.87 vs. 0.60). This indicates that the detection signal arises from how each backbone's internal representation interacts with the scoring function, making architecture selection a security-relevant design choice. To our knowledge, this is the first application of class-conditional conformal prediction to network intrusion detection.**

***Index Terms*—Conformal prediction, intrusion detection, Internet of Things, uncertainty quantification, selective classification, adversarial robustness**

I. Introduction

The rapid expansion of IoT infrastructure—29 billion connected devices projected by 2030 [1]—has made network intrusion detection a pressing problem. Smart home sensors, industrial controllers, medical devices, and autonomous vehicles communicate over heterogeneous protocols (MQTT, CoAP, Zigbee, BLE), generating large volumes of traffic with severe class imbalance: in a typical IoT deployment, benign flows may outnumber any single attack category by three to four orders of magnitude.

Deep learning models now routinely exceed 99% accuracy on IoT traffic benchmarks [2], [3]. In deployment, however, high accuracy on a held-out test set does not translate into reliable operation. SOC analysts face classifiers whose softmax outputs are miscalibrated—a reported 99% confidence may correspond to actual correctness rates well below 90% in out-of-distribution regions [4], [5]. Novel attack types are assigned to the most similar known category without any alert being raised [6], [7]. Adversarial perturbations to flow-level features—achievable by an attacker controlling compromised IoT devices—can cause misclassification rates exceeding 70% [8], [9]. And every input receives a forced prediction, with no mechanism to defer ambiguous cases to a human analyst [10].

Existing uncertainty quantification approaches—MC-Dropout [11], Deep Ensembles [12]—offer approximate solutions but at 10–50$\times$ the inference cost and without finite-sample guarantees on prediction reliability. Conformal prediction (CP) [13], [14] works differently: given a calibration set and a user-chosen significance level α, CP constructs prediction sets guaranteed to contain the true label with probability $\geq 1-\alpha$, under no distributional assumptions beyond exchangeability. The Mondrian extension [13] stratifies calibration by class, ensuring that even attacks constituting 0.06% of traffic receive the target coverage.

Yet conformal prediction has not been applied to IoT intrusion detection with class-conditional guarantees, nor evaluated against adversarial evasion in this setting. We propose CRISP (**C**onformal **R**esilient **I**ntrusion detection with **S**elective **P**rediction). We structure our evaluation around four research questions:

- **RQ1** (*Coverage*): Does Mondrian CP achieve per-class coverage within practical tolerance on IoT traffic, including rare attack classes?
- **RQ2** (*Selective prediction*): Can conformal prediction sets match or surpass heuristic rejection methods while providing formal guarantees?
- **RQ3** (*Adversarial detection*): Can detection be embedded within a coverage-guaranteed framework, and what does this cost relative to standalone detectors?
- **RQ4** (*Architecture dependence*): How does the backbone classifier affect adversarial detection under the same conformal layer?

Our main results are as follows. Mondrian CP achieves per-class coverage within $\pm$0.25% of targets across two datasets, two architectures, and five significance levels (**RQ1**). A unified predict/abstain/alarm framework reaches 99.90% selective accuracy at 92.7% coverage with under 4 μs overhead (**RQ2**). Adversarial detection via a Mondrian-compatible score attains AUC 0.87 against PGD, at a quantified cost of 17–19 AUC

points compared to standalone kNN (**RQ3**), with a complete detection reversal between CNN and attention backbones (**RQ4**).

II. Related Work

Deep learning for IoT IDS. CNN, LSTM, and attention-based architectures dominate the field [3], [15], [16], reaching accuracy above 99% on ACI-IoT-2023 [2] and CIC-IoT-2023 [1]. Ferrag et al. [17] survey over 100 DL-based NIDS. These systems produce point predictions without reliability measures or abstention.

Uncertainty quantification. MC-Dropout [11] and Bayesian CNNs [18] approximate epistemic uncertainty at substantial computational cost. Jorgensen et al. [6] applied UQ to flag unknown encrypted traffic but did not evaluate adversarial scenarios. NeutroSENSE [19] decomposes confidence via neutrosophic logic using shallow ensembles. SelectiveNet [10] learns a rejection function jointly with the classifier but must be retrained for each target coverage level.

Adversarial robustness. IDS vulnerability to gradient-based attacks is well established [9], [20]–[22]. A2PM [23] constrains perturbations to protocol-valid flows. SHAP fingerprinting [24] detects adversarial inputs through attribution profiles. Current defenses tend to be attack-specific or require full retraining, and few provide any formal guarantee on detection rates.

Conformal prediction. Split CP [13], [25] and the APS/RAPS scores [26], [27] have been adopted across domains but remain rare in network security, with recent exceptions in concept drift [28] and telecom anomaly detection [29]. Class-conditional efficiency methods include RC3P [**?**] for rank-calibrated class-wise coverage, clustered CP [**?**] for borrowing strength across similar classes, and conformal risk control [**?**] for group-conditional guarantees under general loss functions. We are not aware of prior work applying Mondrian CP to intrusion detection or jointly evaluating conformal methods across selective classification, adversarial detection, and open-set recognition in a single framework.

III. Methodology

A. Conformal Prediction Preliminaries

Given a trained classifier $\hat{f}{:}\mathcal{X}\to\Delta^K$ with logits $z(x)\in\mathbb{R}^K$, a held-out calibration set $\mathcal{D}_{\text{cal}}=\{(x_i,y_i)\}_{i=1}^n$, and a nonconformity score $s{:}\mathcal{X}\times\mathcal{Y}\to\mathbb{R}$, split CP [25] computes the quantile $\hat{q}=s_{(\lceil(n+1)(1-\alpha)\rceil)}$ and forms prediction sets $\mathcal{C}_\alpha(x)=\{y : s(x,y)\le\hat{q}\}$. Under exchangeability between calibration and test data, the coverage guarantee

$$P(y_{\text{test}}\in\mathcal{C}_\alpha(x_{\text{test}}))\ge 1-\alpha \tag{1}$$

holds exactly in finite samples [14]. The ceiling formula—rather than interpolated quantile estimation—is essential for this finite-sample property [13]. Mondrian CP stratifies calibration by class, yielding per-class quantiles $\hat{q}_k$ and the class-conditional guarantee:

$$P\big(y\in\mathcal{C}_\alpha^M(x)\mid y{=}k\big)\ge 1-\alpha,\quad \forall\, k\in\mathcal{Y} \tag{2}$$

For imbalanced IoT traffic, this prevents marginal coverage from being satisfied at the expense of rare classes. With 11 classes spanning three orders of magnitude in prevalence, marginal CP could in principle achieve 90% overall coverage while covering the rarest class at well below 50%.

B. CRISP Architecture

CRISP is applied after training and requires no changes to the base model. It takes softmax outputs, logits, and a held-out calibration set as input, and operates through three modules (Fig. 1).

Module 1: Dual scoring. From a single forward pass, two nonconformity scores are computed in parallel for different operational purposes. The Adaptive Prediction Sets (APS) score [26],

$$s_{\text{APS}}(x,y)=\sum_{j=1}^{L(y)}\hat{f}(x)_{\pi_j}-(1{-}u)\cdot\hat{f}(x)_y \tag{3}$$

sorts softmax outputs in descending order and accumulates probability mass until the true class y is included. Here π is the descending-probability permutation, $L(y)$ is the rank of class y, and $u{\sim}U(0,1)$ provides randomized smoothing for finite-sample validity[1]. When the model assigns high probability to the true class, s_{APS} is small and the resulting prediction set is typically a singleton. When the model is uncertain, more classes accumulate before y is reached, producing larger sets.

The LogSoftmax-T score targets adversarial sensitivity:

$$s_{\text{LST}}(x,y)=-T\cdot\log\frac{e^{z_y(x)/T}}{\sum_k e^{z_k(x)/T}} \tag{4}$$

It operates on raw logits at temperature $T{=}1$ (selected by sweep over $\{0.5,1,2,5\}$; see Section V-C). The key constraint is that s_{LST} depends on the hypothesized class y—unlike the label-free energy score $E(x){=}{-}T\log\sum_k e^{z_k/T}$ [30], which assigns the same value regardless of y. In our experiments, label-free scores under Mondrian calibration produced degenerate per-class quantiles, resulting in approximately 95% false alarm rates. The class-conditional structure of s_{LST} avoids this collapse while remaining sensitive to logit-space distortions caused by adversarial perturbation.

We also evaluated RAPS [27] ($\lambda{=}0.01$, $k_{\text{reg}}{=}5$); on ACI-IoT-2023, RAPS produced results identical to APS because the base classifiers rank the true class first for over 99% of test samples. The RAPS regularization penalty effectively vanishes on data where the base model is already well-calibrated at the top of its ranking.

Module 2: Mondrian conformal layer. Class-specific quantiles $\hat{q}_k$ are estimated from the APS scores of calibration samples belonging to class k. At inference, the APS-based prediction set cardinality $|\mathcal{C}_\alpha^M(x)|$ drives the primary operational

[1] We implement u via SHA256 hashing of model outputs for reproducibility; collision probability $<10^{-30}$ on `float32` vectors. Without smoothing, the non-randomized ceiling quantile $\hat{q}=s_{(\lceil(n+1)(1-\alpha)\rceil)}$ already provides valid (conservative) coverage [13]; our hash-based u tightens the bound while producing empirical coverage within $\pm0.25\%$ of the target, consistent with the conservative baseline.

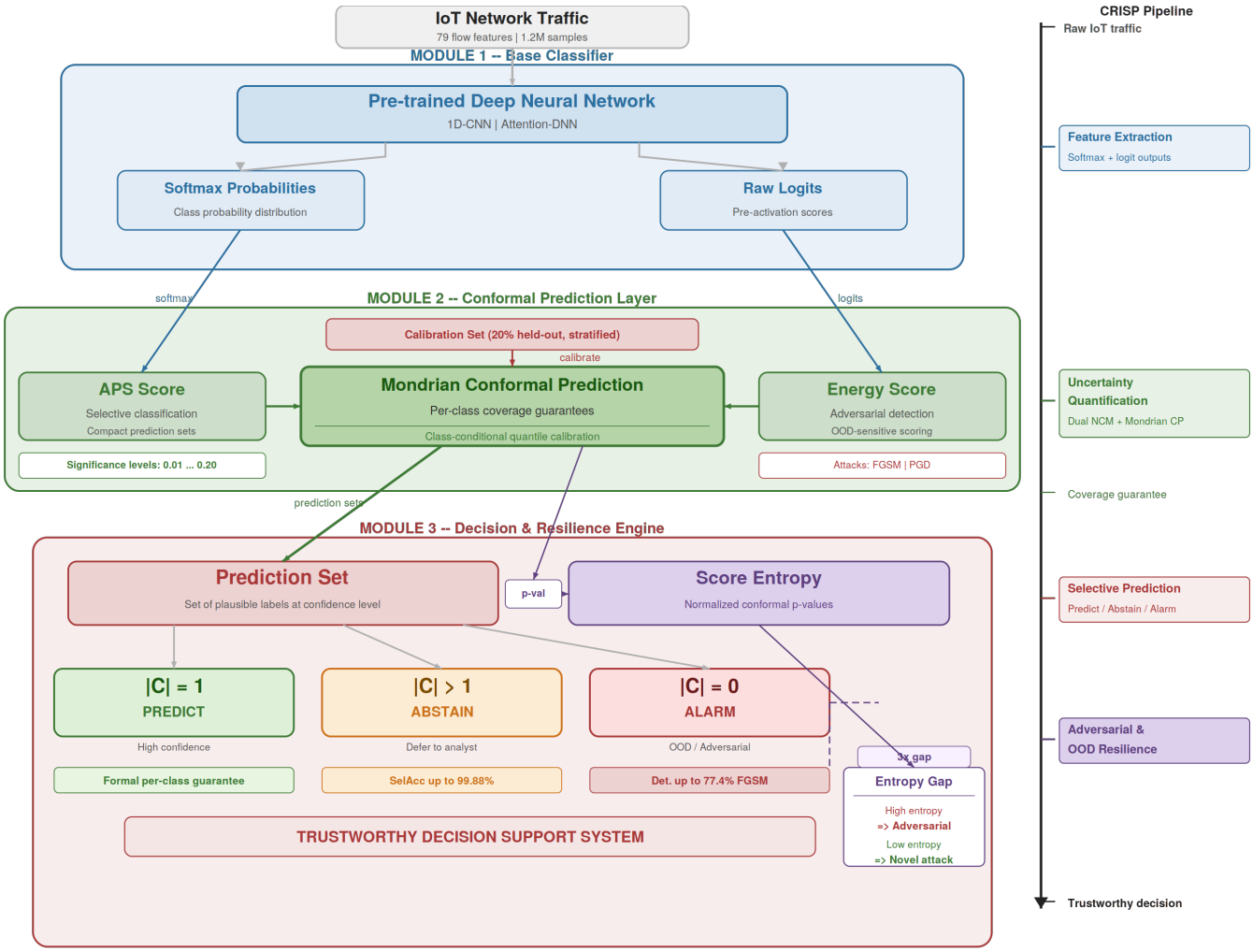

Fig. 1. Overview of the CRISP pipeline. Module 1 extracts softmax and logit representations from the base classifier. Module 2 computes APS and LogSoftmax-T nonconformity scores with Mondrian calibration. Module 3 maps set cardinality to operational decisions; conformal entropy discriminates adversarial from novel inputs.

decision: $|\mathcal{C}|{=}1$ triggers a confident prediction forwarded for automated response; $|\mathcal{C}|{>}1$ signals ambiguity and defers the input to a Tier-1 analyst; $|\mathcal{C}|{=}0$ indicates that no class is consistent with the input at the required confidence level and escalates to Tier-2 investigation. When $|\mathcal{C}|{=}0$, Module 3 is invoked to determine whether the anomaly is adversarial or novel. The selective accuracy and coverage are:

$$\text{SelAcc} = \frac{\sum_i \mathbf{1}[\hat{y}_i{=}y_i \wedge |\mathcal{C}|{=}1]}{\sum_i \mathbf{1}[|\mathcal{C}|{=}1]}, \quad \text{SelCov} = \frac{\sum_i \mathbf{1}[|\mathcal{C}|{=}1]}{N} \tag{5}$$

Module 3: Resilience engine. A second Mondrian CP instance, calibrated with LogSoftmax-T on the same calibration set $\mathcal{D}_{\text{cal}}$, produces per-class conformal p-values:

$$p_k(x) = \frac{|\{i \in \mathcal{D}_{\text{cal}}^{(k)} : s_i \geq s_{\text{LST}}(x,k)\}| + 1}{n_k + 1} \tag{6}$$

Reusing $\mathcal{D}_{\text{cal}}$ for both APS and LogSoftmax-T calibration is valid because the two scores operate on different representations (softmax probabilities vs. temperature-scaled logits) and calibrate independent quantile sets; no information leaks from one calibration to the other. An input is flagged as anomalous when $\max_k p_k(x) < \epsilon$, where ϵ is set to the empirical α-quantile of $\max_k p_k$ on clean calibration data. This threshold is not multiplicity-corrected across the K classes; applying Bonferroni (ϵ/K) would reduce false alarms but also detection sensitivity. We report empirical false alarm rates on clean test data directly. To distinguish adversarial inputs—uniformly nonconforming across all classes—from novel attacks that conform to some classes but not others, we compute the normalized entropy $H_s(x) = -\sum_k \tilde{p}_k \log \tilde{p}_k$ where $\tilde{p}_k = p_k / \sum_j p_j$.

Coverage guarantees hold under exchangeability. When that assumption is violated—as with adversarial inputs or distribution drift—coverage is not assured. The conformal framework provides the detection mechanism; detection rates are measured empirically.

TABLE I
ACI-IoT-2023 CLASS DISTRIBUTION.

Family	Class	n	%
Legitimate	Benign	329,295	26.7
Volumetric	ICMP Flood	225,234	18.3
	DNS Flood	46,935	3.8
	SYN Flood	13,857	1.1
	UDP Flood	791	0.06
Recon.	Port Scan	441,282	35.8
	Ping Sweep	71,928	5.8
	OS Scan	37,524	3.0
	Vuln. Scan	39,537	3.2
Credential	Dict. Attack	6,380	0.5
Application	Slowloris	18,643	1.5

TABLE II
COVERAGE ON ACI-IOT-2023 (246K TEST) AND CICIDS2017 (427K TEST). APSS SHOWN AS ACI/CIC.

		1D-CNN			Attn-DNN		
α	Tgt	ACI (%)	CIC (%)	APSS	ACI (%)	CIC (%)	APSS
0.01	99.0	98.98	99.01	2.01/1.08	99.02	99.01	1.06/1.14
0.05	95.0	94.86	95.04	0.97/1.00	95.08	94.98	0.98/1.02
0.10	90.0	89.85	90.04	0.91/0.94	90.13	89.98	0.92/0.96
0.15	85.0	84.89	84.99	0.86/0.88	85.18	84.96	0.86/0.90
0.20	80.0	79.98	80.01	0.81/0.83	80.25	79.88	0.81/0.85

IV. EXPERIMENTAL SETUP

A. Datasets

We evaluate on two benchmarks with different characteristics. ACI-IoT-2023 [2] contains 1,231,411 flows with 79 features across 12 traffic classes in a smart home IoT environment. ARP Spoofing (n=5) is allocated exclusively to the training split; the remaining 11 classes span four threat families (Table I). Class imbalance spans three orders of magnitude, from Port Scan (441K flows, 35.8%) to UDP Flood (791 flows, 0.06%). CICIDS2017 [31] provides 2,136,413 flows with 78 features and 8 classes after filtering rare categories (n<50). The imbalance is more severe: 85.7% benign, with the rarest class (XSS) at 0.03%.

Data is split into four disjoint stratified subsets: training (50%), validation (10%), calibration (20%), and test (20%). The validation set is used only for early stopping. The calibration set is reserved for conformal quantile estimation and is never exposed during model training or selection. Feature scaling (`StandardScaler`) is fit on training data only and applied to all splits to prevent leakage.

B. Base Classifiers

Two architecturally distinct models are used to test model-agnosticism. A **1D-CNN** (265K parameters) applies three Conv1D–BatchNorm–ReLU–MaxPool blocks with 64, 128, and 256 filters, followed by adaptive average pooling and two dense layers (128, 64 units) with dropout (0.3). An **Attention-DNN** (85K parameters) projects the 79 features into 64-dimensional tokens via chunking (chunk size 8, yielding 10 tokens), processes them through a 2-layer transformer encoder with 4 attention heads and GELU activation, and applies mean pooling before two dense layers. Both are trained with Adam [32] (lr= 10^{-3}, weight decay 10^{-5}) and early stopping (patience 10) on a single NVIDIA RTX 4070 GPU (12 GB VRAM, CUDA 12.1, PyTorch 2.1). The CNN reaches 99.55% accuracy (F1-macro 0.970) at epoch 73; the Attention-DNN reaches 99.45% (F1-macro 0.955) at epoch 100 with sharper softmax distributions. Conformal inference overhead—K score evaluations and K quantile comparisons per sample—is measured at under 4 μs on the same hardware (averaged over 246K test samples, excluding the DNN forward pass).

C. Threat Model and Baselines

We model a post-deployment evasion adversary who controls compromised IoT devices and can modify flow-level features (packet sizes, timing statistics, flag distributions) within an L_∞ budget ϵ in standardized feature space, without access to the model, its calibration data, or the conformal layer. The complementary threat—perturbing benign traffic to trigger false alarms (an availability attack on the IDS)—is a distinct threat model that would primarily stress the false alarm rate of the conformal alarm pathway; it is not evaluated here but would be an informative stress test. FGSM [20] ($\epsilon\in\{0.01, 0.05, 0.10\}$) and PGD [21] ($\epsilon$=0.05, 20 steps, step size 0.01) are applied to attack-class samples only; benign samples remain unperturbed. These perturbations are unconstrained in feature space and may produce protocol-invalid flows; reported detection rates should be interpreted as upper bounds. For open-set evaluation, UDP Flood—the rarest attack and spectrally closest to known flood types—is withheld from all training and calibration splits.

Selective classification baselines include softmax thresholding (τ=0.95), MC-Dropout [11] (50 passes, threshold at the 80th calibration-set entropy percentile), and standard (non-Mondrian) CP. For adversarial detection, we compare against six standalone methods: kNN on logits (k=5), Mahalanobis distance on calibration-set logits, Isolation Forest on softmax probabilities (n=200), MSP [5], and ODIN with $T\in\{100, 1000\}$.

V. RESULTS

A. Coverage Validation (RQ1)

Table II reports the coverage results. Across both datasets, all five α levels, and both architectures, the deviation between empirical and target marginal coverage stays within $\pm$0.25% on ACI-IoT-2023 and $\pm$0.12% on CICIDS2017. On CICIDS2017, the same conformal code was applied without any dataset-specific tuning—the only change was the base classifier training, which used inverse-frequency class weighting to handle the 85.7% benign ratio.

Per-class coverage (Table III, Fig. 2b) holds within $\pm$1.8% for every class with more than 1000 calibration samples.

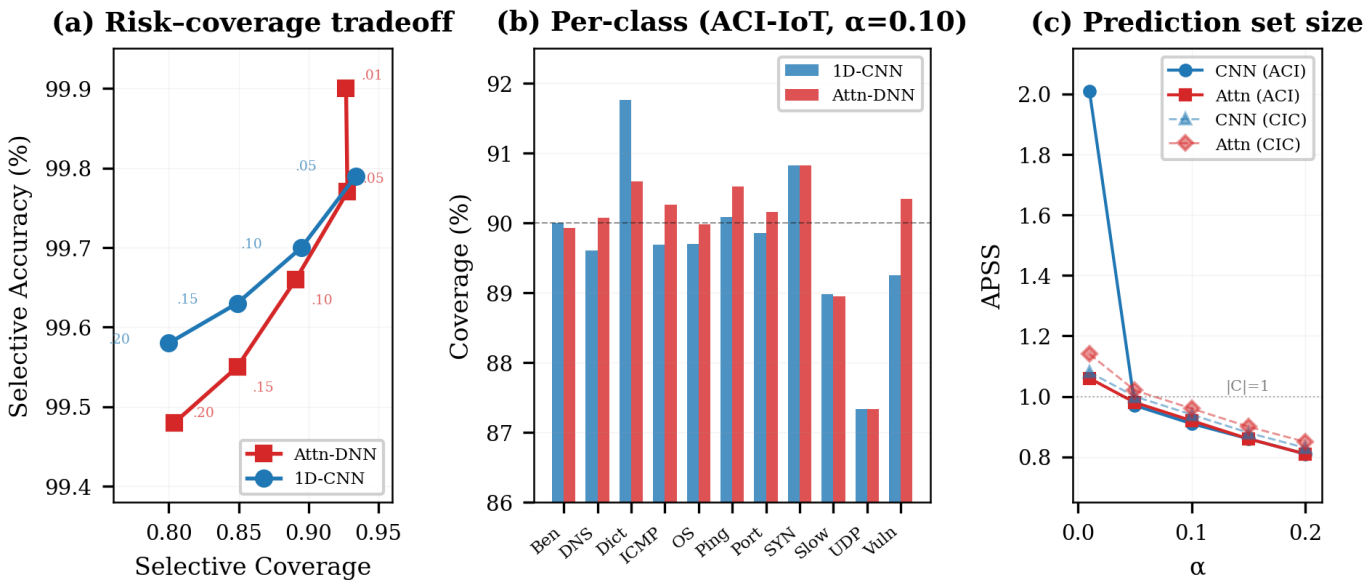

Fig. 2. (a) Risk–coverage tradeoff: CRISP traces a Pareto curve as α varies, dominating the unrejected DNN at each operating point; MC-Dropout reaches 100% accuracy but at fixed 80% coverage without guarantees. (b) Per-class Mondrian coverage at α=0.10: all classes within $\pm$1.8% of the 90% target except UDP Flood (n=158, $-$2.7%). (c) Average Prediction Set Size across α and both datasets.

TABLE III
PER-CLASS COVERAGE (%) AT α=0.10 ON ACI-IOT-2023.

Class	n (test)	CNN	Attn	$\Delta_{\max}$
Benign	65,859	90.00	89.93	-0.07
DNS Flood	9,387	89.61	90.08	-0.39
Dict. Atk	1,276	91.77	90.60	$+1.77$
ICMP Flood	45,047	89.69	90.26	-0.31
OS Scan	7,505	89.70	89.98	-0.30
Ping Sweep	14,386	90.09	90.52	$+0.52$
Port Scan	88,257	89.86	90.16	-0.14
SYN Flood	2,771	90.83	90.83	$+0.83$
Slowloris	3,729	88.98	88.95	-1.05
UDP Flood	158	**87.34**	**87.34**	-2.66
Vuln. Scan	7,907	89.25	90.35	-0.75

TABLE IV
SELECTIVE CLASSIFICATION ON CLEAN ACI-IOT-2023 DATA.

Method	Model	SelAcc (%)	SelCov (%)	Guar.
DNN (no rejection)	Attn	99.45	100	✗
Softmax τ=0.95	Attn	99.90	96.8	✗
MC-Dropout	Attn	**100.0**	80.0	✗
Standard CP α=0.10	Attn	99.65	89.3	marginal
CRISP α=0.10	Attn	99.66	89.1	✓
CRISP α=0.01	Attn	**99.90**	92.7	✓

UDP Flood, with $n_{\text{cal}}\approx$158, under-covers by 2.7%—the same deviation for both models, which is consistent with finite-sample quantile estimation variance rather than a model-specific deficiency. For all classes with $n_{\text{test}} > 1000$, the worst deviation is 1.05% (Slowloris). As a rough guideline, per-class quantile estimation requires $n_k \gtrsim 1/(1-\alpha)$ samples for the ceiling formula to be well-defined; for α=0.10, this gives $n_k \geq 10$. Empirically, we observe stable coverage ($\Delta < 1\%$) for $n_k > 1000$ and increasing variance below that threshold.

At α=0.01, the two architectures diverge. The Attention-DNN produces near-singleton sets (APSS=1.06) with 99.90% selective accuracy at 92.7% coverage, while the CNN yields APSS=2.01—predominantly two-class sets with poor singleton accuracy. The divergence traces to softmax geometry: self-attention learns sharper logit separations, requiring less cumulative mass to reach the true class under APS. For $\alpha \geq 0.05$, both models produce APSS<1 and selective accuracy above 99.5%. Values below 1.0 arise from the mixture of singletons, empty sets, and occasional multi-class sets—at α=0.05, the CNN produces 93.4% singletons, 4.9% empty sets, and 1.6% multi-class sets. The risk–coverage tradeoff is shown in Fig. 2a.

B. Selective Classification (RQ2)

Table IV compares selective methods on clean data. MC-Dropout reaches 100% accuracy but at a fixed 80% coverage determined by its entropy threshold, with no formal guarantee and 50$\times$ inference cost ($\sim$200 μs per sample for 50 stochastic passes on our RTX 4070, vs. <4 μs for the conformal layer). CRISP at α=0.01 reaches the same accuracy level (99.90%) while covering 92.7% of traffic—13 percentage points more—with a formal per-class guarantee. Standard (non-Mondrian) CP achieves comparable selective accuracy to CRISP but provides only marginal coverage, which may fail to protect rare classes.

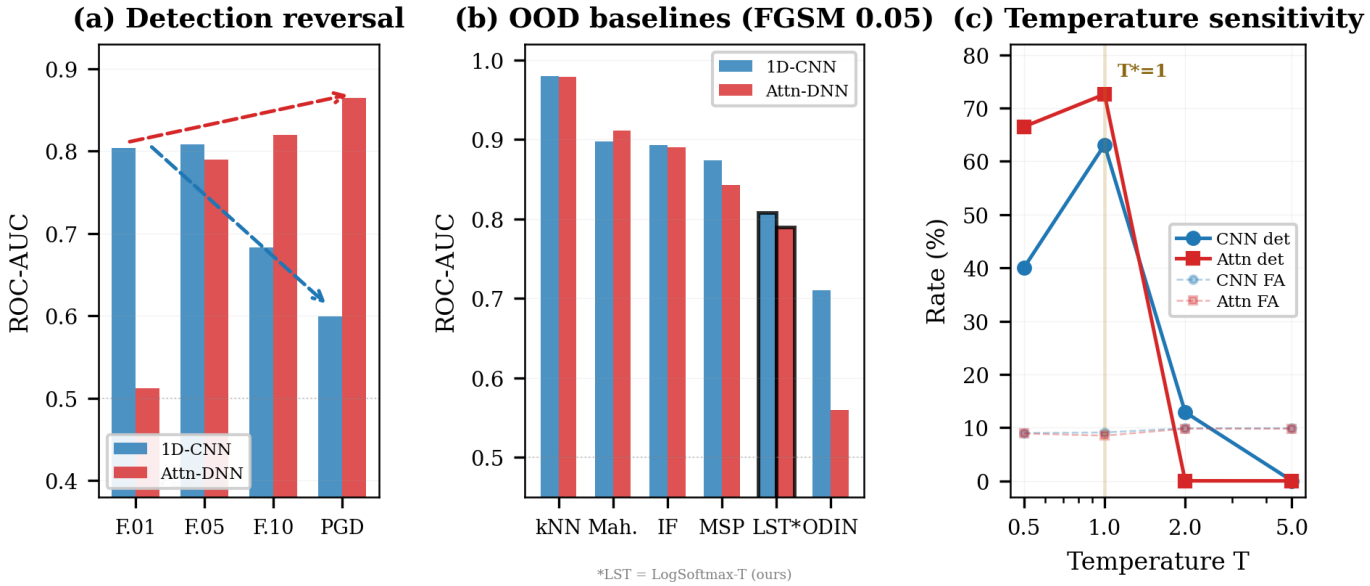

Fig. 3. (a) Detection reversal across architectures (RQ4): the CNN dominates under weak FGSM while the Attention-DNN prevails under PGD. Dashed arrows trace each model's AUC trajectory. (b) Standalone detector comparison on FGSM ϵ=0.05 (RQ3): kNN-logits at 0.98 but not Mondrian-compatible; LogSoftmax-T (black border) is the strongest compatible score. (c) Temperature sensitivity: T=1 is optimal; detection collapses at $T\geq2$ while false alarm remains near 9%.

TABLE V
STANDALONE OOD DETECTION COMPARISON (FGSM ϵ=0.05).

Method	CNN	Attn	Mond.	Guar.
kNN logits (k=5)	**0.980**	**0.979**	✗	none
Mahalanobis	0.898	0.911	✗	none
IF softmax (n=200)	0.893	0.890	✗	none
MSP	0.874	0.843	✗	none
LogSoftmax-T (CRISP)	0.808	0.790	✓	coverage
ODIN (T=100)	0.710	0.559	✗	none

C. Adversarial Detection (RQ3) and Architecture Dependence (RQ4)

Table V and Fig. 3b compare detection methods. kNN on logits leads at 0.980/0.979, measuring mean Euclidean distance to the five nearest calibration neighbors in the 12-dimensional logit space. Mahalanobis (0.898/0.911) and Isolation Forest on softmax (0.893/0.890) follow. All three are label-free: they produce a single score $s(x)$ regardless of the hypothesized class, which makes them incompatible with Mondrian calibration's requirement for class-conditional scores $s(x, y)$. LogSoftmax-T is the strongest Mondrian-compatible alternative, at 0.808/0.790. The 17–19 point gap against kNN reflects the structural price of class-conditional scoring.

On RQ4, Fig. 3a shows a reversal in detection behavior between the two backbones. Under weak perturbation (FGSM ϵ=0.01), the CNN detects well (AUC 0.80) while the Attention-DNN is near chance (0.51). Under PGD, the Attention-DNN reaches 0.87 and the CNN drops to 0.60. Weak FGSM barely degrades the Attention-DNN's predictions—self-attention absorbs small feature perturbations without significant logit distortion—so its logits remain close to the clean distribution and escape detection. The CNN's less smooth decision boundary produces measurably distorted logits even under weak attack. PGD inverts this pattern: the CNN's flatter softmax geometry permits adversarial examples that closely mimic clean data, while the Attention-DNN's sharper logit structure produces distinct adversarial artifacts that LogSoftmax-T picks up.

A temperature sweep (Fig. 3c) confirms T=1 as optimal for both architectures. At T=0.5, detection is lower (40% CNN, 67% Attn) because the sharpened softmax suppresses score variation. At $T\geq2$, detection collapses near zero while false alarm rates remain stable around 9%, confirming that the detection–false alarm tradeoff is controlled by score sensitivity to logit perturbations, not by threshold calibration.

D. Open-Set, Ablation, and Cross-Dataset Generalization

With UDP Flood withheld, conformal entropy separates novel from known attack traffic by 2.6× (CNN) and 3.1× (Attention-DNN). Empty-set alarm rates on the novel class are modest (4–5%), which is expected given the spectral similarity between UDP Flood and known flood variants. We note that AUROC/AUPRC for the entropy-based novel-vs-known discrimination are not reported here; the entropy gap is presented as a signal characterization rather than a calibrated detector. Operationalizing this signal—threshold selection, precision-recall characterization, and integration with the existing alarm pathway—is left for future work.

Standard and Mondrian CP produce comparable selective accuracy (99.65–99.70%), confirming that per-class calibration does not degrade clean-data performance. Switching from APS

to LogSoftmax-T increases adversarial alarm rates by roughly an order of magnitude (from 6–8% to 63–73%), with the Attention-DNN benefiting more (72.6% vs. 63.1%), consistent with the architecture-dependent pattern described above.

On CICIDS2017, coverage deviates by at most $\pm 0.12\%$—tighter than ACI-IoT-2023—using identical conformal code. Selective accuracy exceeds the base classifier's raw performance at every α level through principled abstention. The conformal layer requires no dataset-specific modification to transfer across traffic distributions, class counts, and imbalance ratios.

VI. Discussion

Coverage validity (RQ1). The deviation between empirical and target coverage—at most $\pm 0.25\%$ across twenty measurements—follows from the conformal construction and is not an empirical coincidence. What the experiments verify is that the implementation is correct and that exchangeability holds to a sufficient approximation on these benchmarks. Under adversarial perturbation, exchangeability breaks and coverage is no longer assured. The detection capability that CRISP provides in that regime is empirical, and the remaining questions assess its effectiveness.

Selective prediction (RQ2). CRISP at α=0.01 matches MC-Dropout's accuracy (99.90%) at 13 percentage points more coverage, with a formal per-class guarantee instead of a heuristic threshold. In absolute terms, MC-Dropout's 50 forward passes add $\sim$200 μs per sample on our hardware, compared to CRISP's $<$4 μs conformal overhead—a 50$\times$ gap that compounds at IoT-scale throughput. The 7% of inputs deferred at α=0.01 represents traffic the model cannot classify with the required confidence—a fraction the operator controls through α. In a SOC workflow, this maps to the Tier-1 analyst queue, and its size can be tuned to match staffing capacity. We note that efficiency-focused CP methods such as RC3P [**?**] and clustered CP [**?**] could further reduce prediction set sizes for rare classes, potentially recovering some of the coverage lost to finite-sample effects on small n_k; these are orthogonal improvements that could be layered onto CRISP without modifying the detection pipeline.

Integration cost (RQ3). The 17–19 AUC point gap between LogSoftmax-T and kNN-on-logits is the main tradeoff in this work. It arises because Mondrian calibration demands a class-conditional score, excluding label-free methods that can exploit the full logit space. In settings where formal coverage, abstention, and detection must coexist in a single pipeline, no standalone detector offers this. Where detection alone is the priority, kNN-on-logits at 0.98 AUC remains the stronger option, but forfeits the coverage guarantee and the predict/abstain/alarm decision structure. A promising direction to narrow this gap is constructing class-conditional kNN distances as nonconformity measures: for each hypothesized class k, the score would be the distance to the k-th nearest neighbor among calibration samples of class k. This preserves the Mondrian-compatible $s(x, y)$ structure while retaining the geometric sensitivity that gives kNN its detection advantage. Recent work on local density corrections and bagged k-distances for OOD detection [33] suggests that such class-conditional variants are feasible, though their calibration cost scales with $K \times n_{\text{cal}}$.

Architecture effects (RQ4). The detection reversal has direct implications for deployment. An attention-based backbone provides stronger defense against iterative adversaries (PGD: AUC 0.87), while a convolutional backbone responds more sharply to low-budget, single-step perturbations (FGSM ϵ=0.01: AUC 0.80). This means that model selection is itself a security decision, even within a framework designed to be model-agnostic. A dual-backbone deployment—with CRISP wrapping both architectures in parallel—could cover both perturbation regimes.

VII. Conclusion

CRISP applies Mondrian conformal prediction as a post-hoc wrapper to equip any trained IoT classifier with per-class coverage guarantees, selective prediction, and integrated adversarial detection. Per-class coverage holds within $\pm 0.25\%$ of theoretical targets across all configurations tested (**RQ1**). Selective accuracy reaches 99.9% at 92.7% coverage, exceeding heuristic baselines while carrying formal guarantees (**RQ2**). Adversarial detection is feasible within the guaranteed framework, at a measured cost of 17–19 AUC points relative to standalone detectors (**RQ3**). The choice of backbone has a substantial, previously unreported effect on detection effectiveness (**RQ4**).

Several limitations should be acknowledged. The evaluation uses randomized stratified splits; time-ordered or device-aware partitioning would better approximate deployment conditions where flows from the same device or time window are correlated. Adversarial perturbations are unconstrained in feature space and may not produce protocol-valid flows; evaluation under constrained attacks such as A2PM [23] is needed to establish detection rates on realistic adversarial traffic. The architecture-dependent reversal is observed at a single random seed; multi-seed replication, ideally including additional backbones (e.g., BiLSTM, MLP), would strengthen the generality of this finding. The open-set evaluation reports entropy separation ratios but not AUROC/AUPRC for the novel-class discrimination task; these metrics would be needed before the entropy signal can be deployed as a standalone detector.

Future work includes online CP for concept drift [28], improved rare-class set efficiency through RC3P [**?**] or clustered CP [**?**], and Mondrian-compatible scores based on class-conditional distances to close the gap with standalone detectors.

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